



**SBME
UBC**

Synthetic Biology

BMEG 426

Kalaimaran Balasothy

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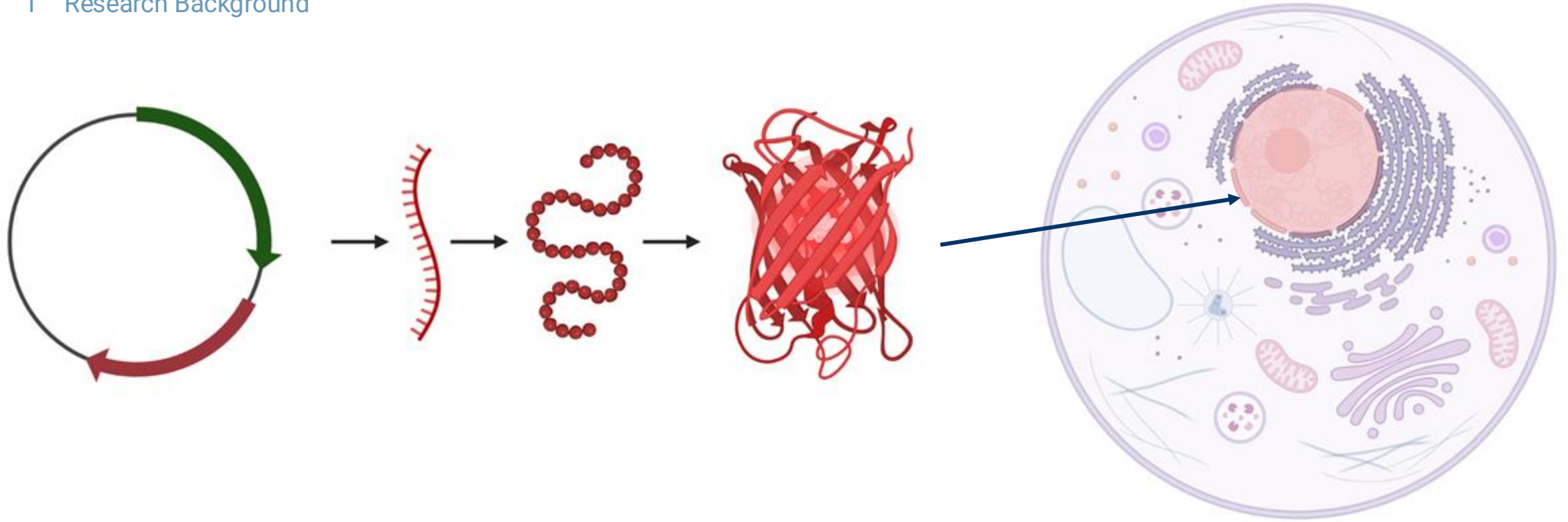
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Introduction

- Cells are simply complex computers that we're slowly learning how to program
- DNA is the fundamental unit of life → One that we are now slowly being able to write
- The experiment in this course is focused on producing a genetic device & testing it via injecting it in a mammalian cell.

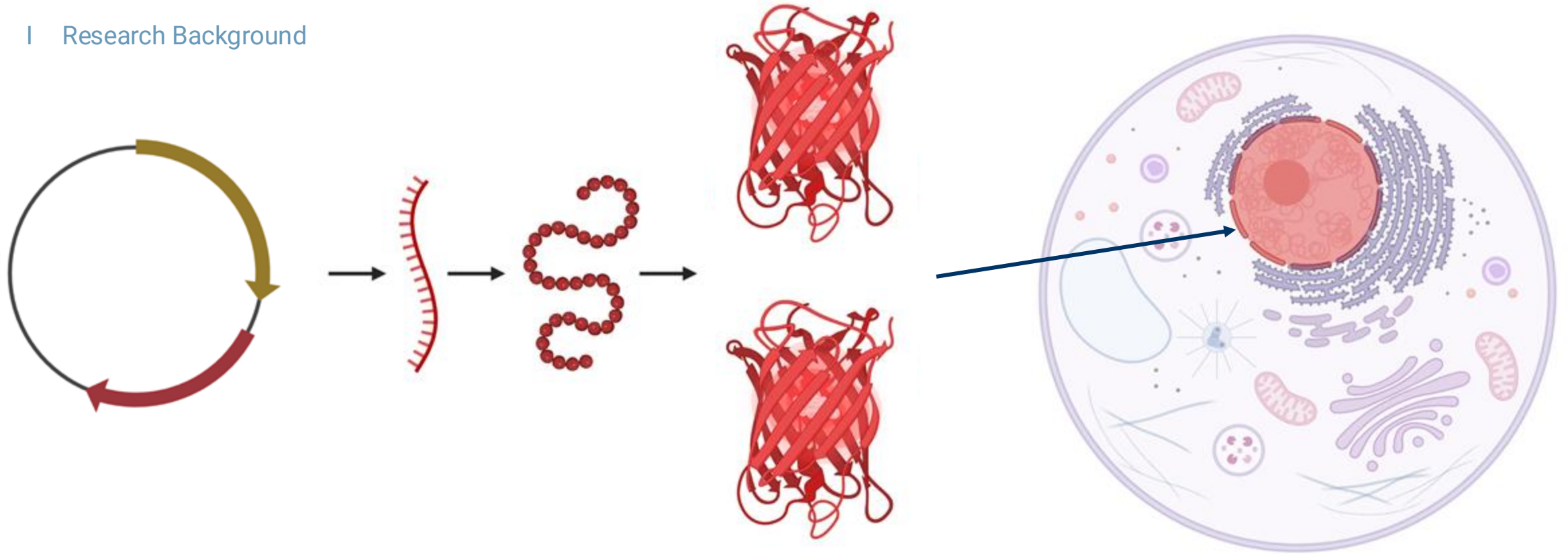


I Research Background



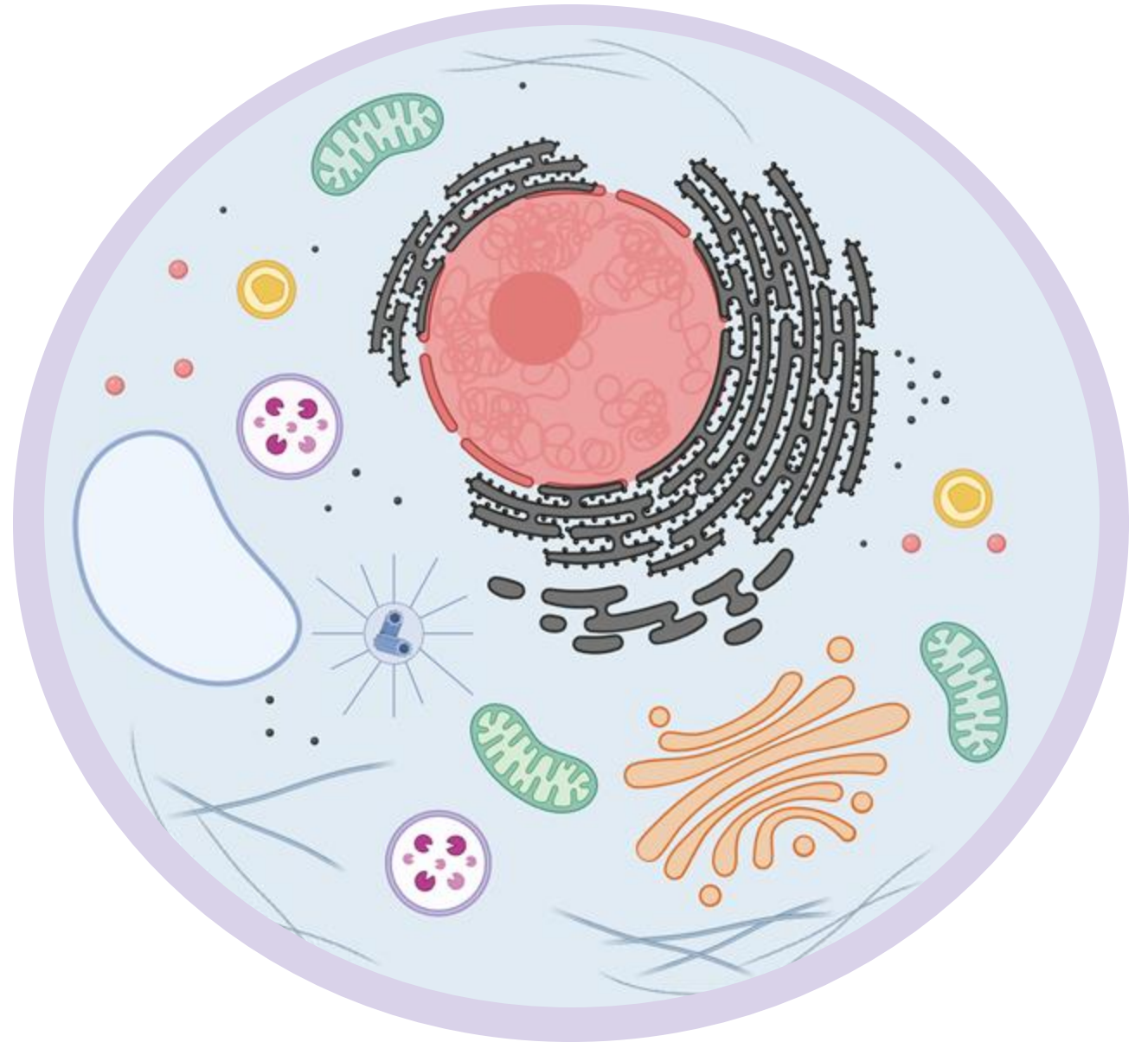
- A DNA plasmid consisting of crucially a promoter & gene for both a subcellular tag & fluorescent protein
- The cell will produce the protein which will then tag a particular location (eg. the nucleus).

I Research Background



- A DNA plasmid consisting of crucially a promoter & gene for both a subcellular tag & fluorescent protein
- The cell will produce the protein which will then tag a particular location (eg. the nucleus).
- By changing the constitutively active promoter we can identify if it changes expression (eg. increases production so brighter fluorescence)

The end goal is to create a catalog of genetic devices where each device **uniquely tags a subcellular region with a unique colour** → a cellular painting in other words.

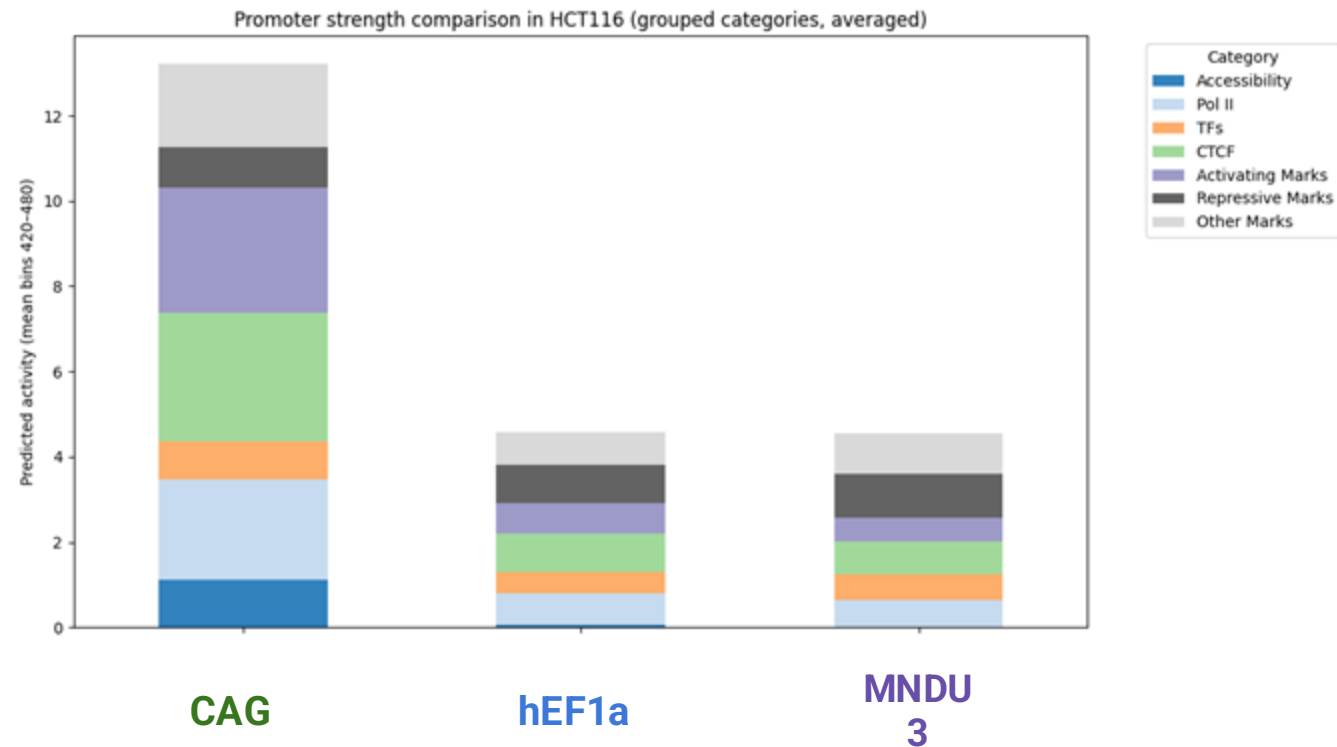
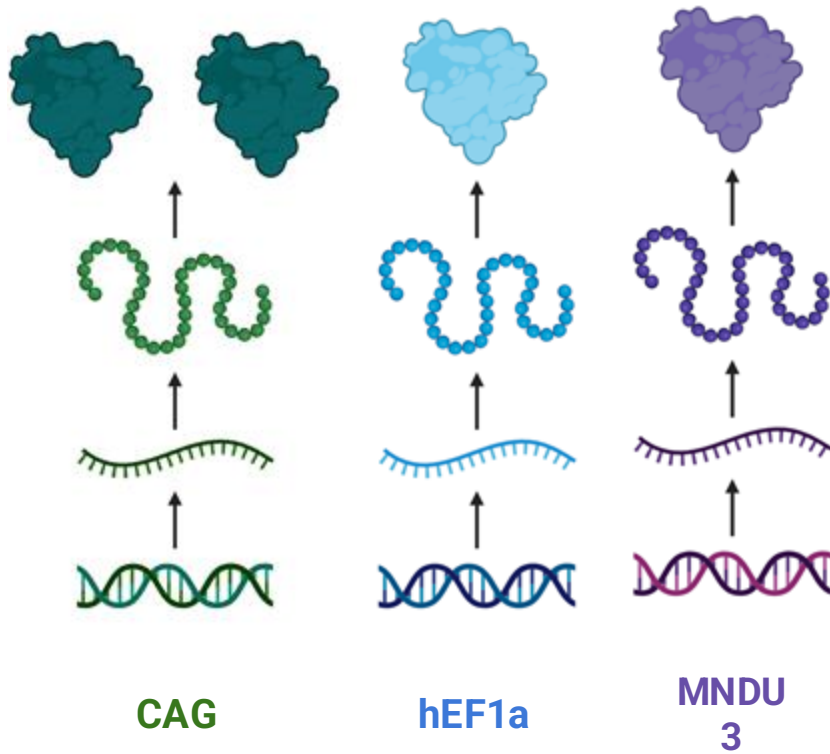


Central Question

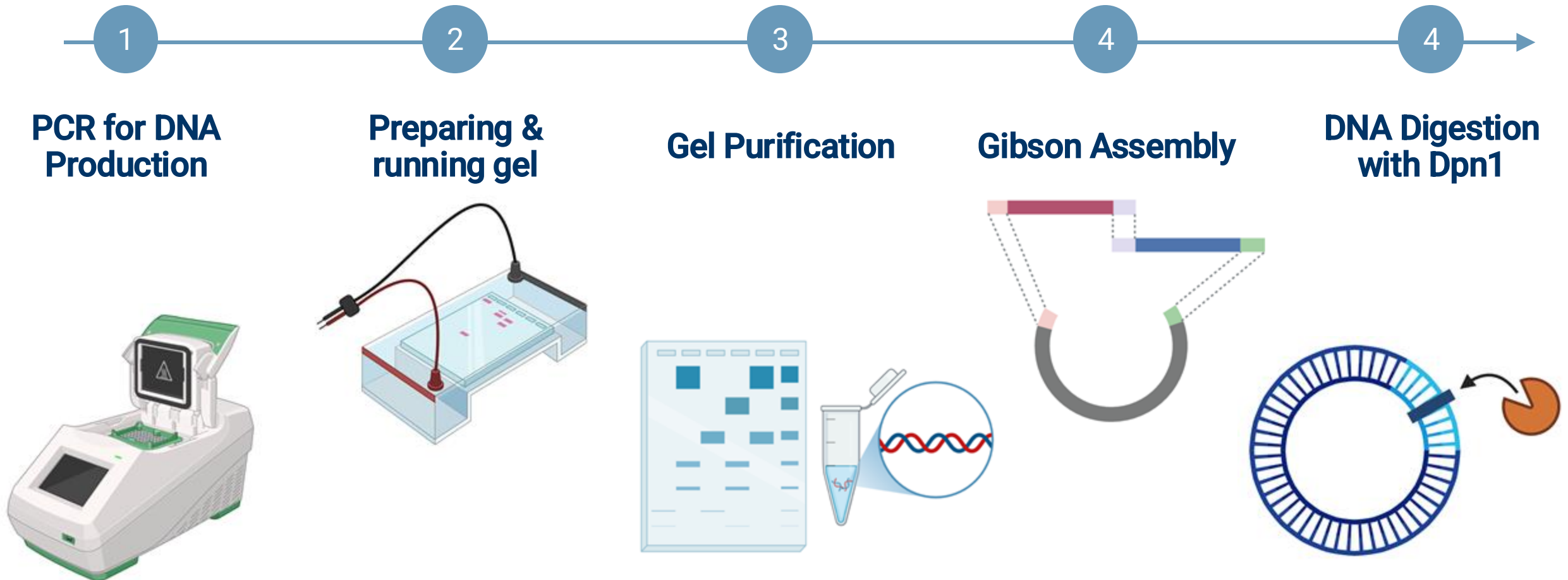
How do **different constitutively active promoters** impact the level of expression of our protein of interest?

Hypothesis

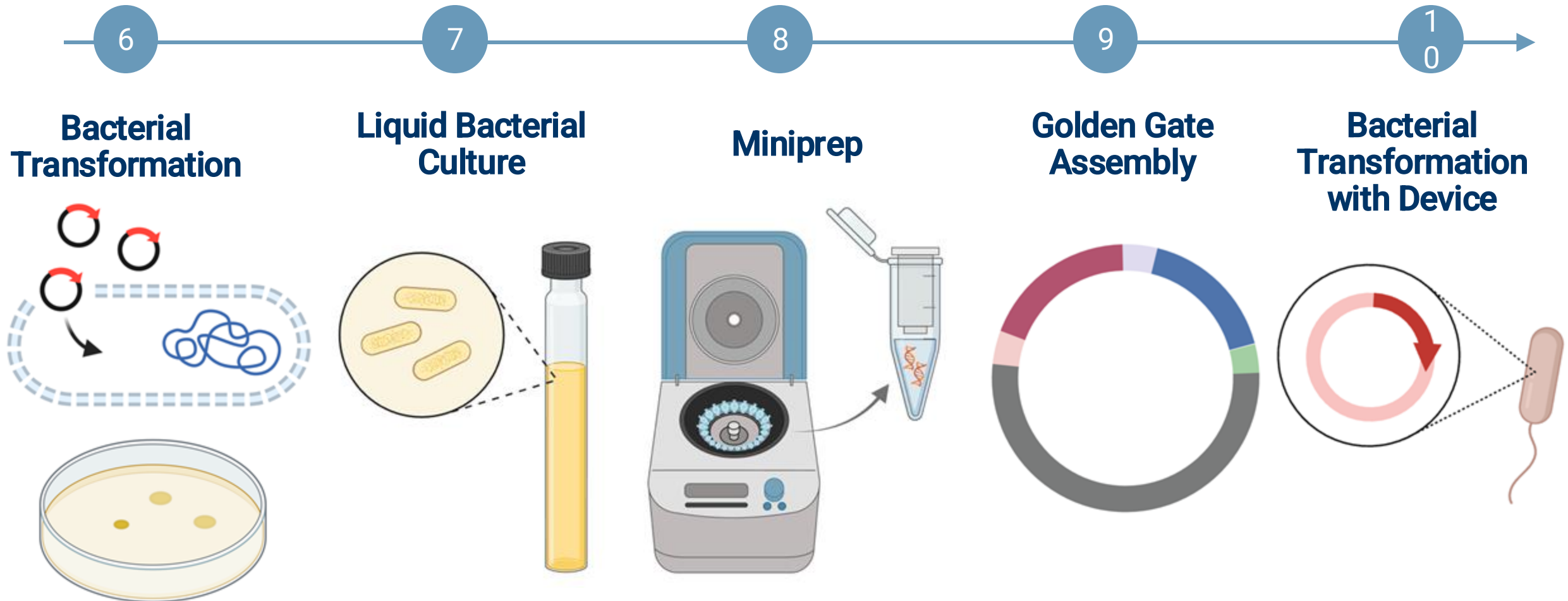
Using *Enformer*, I hypothesize that CAG will lead to the highest expression with hEF1a & MNDU3 having similar levels



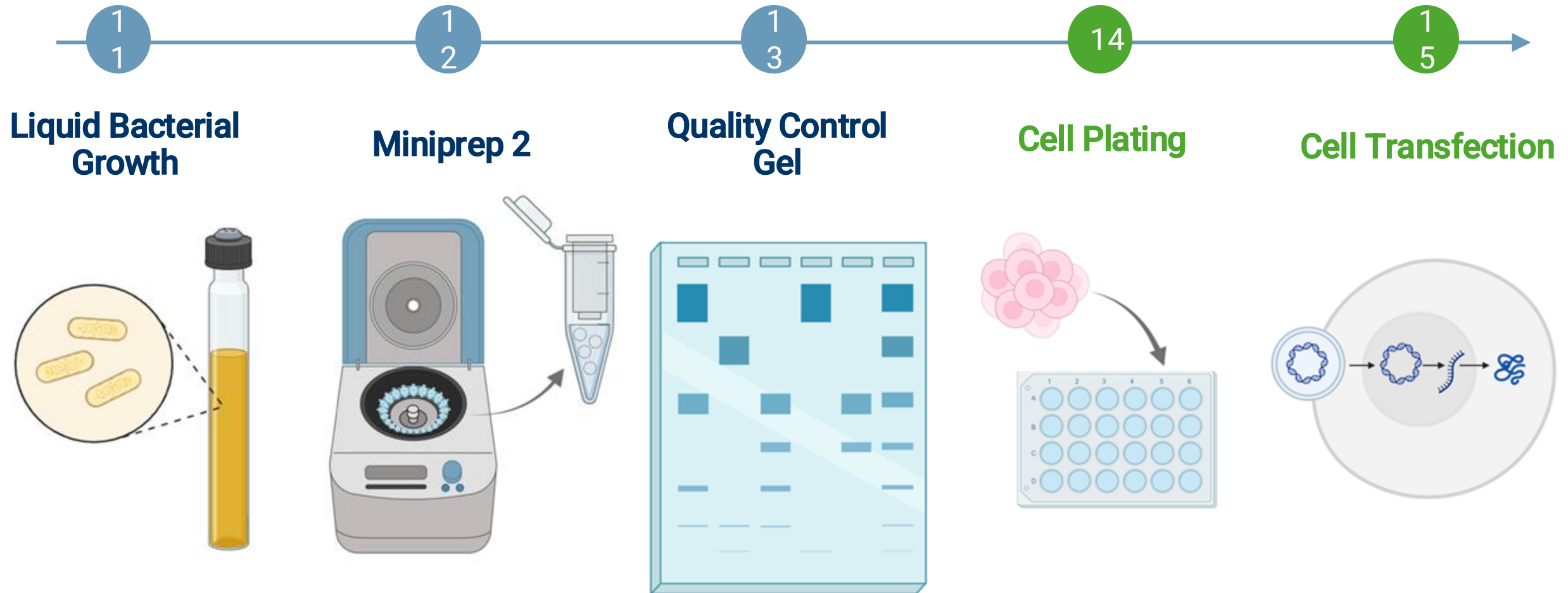
Experiment Timeline (1/3)



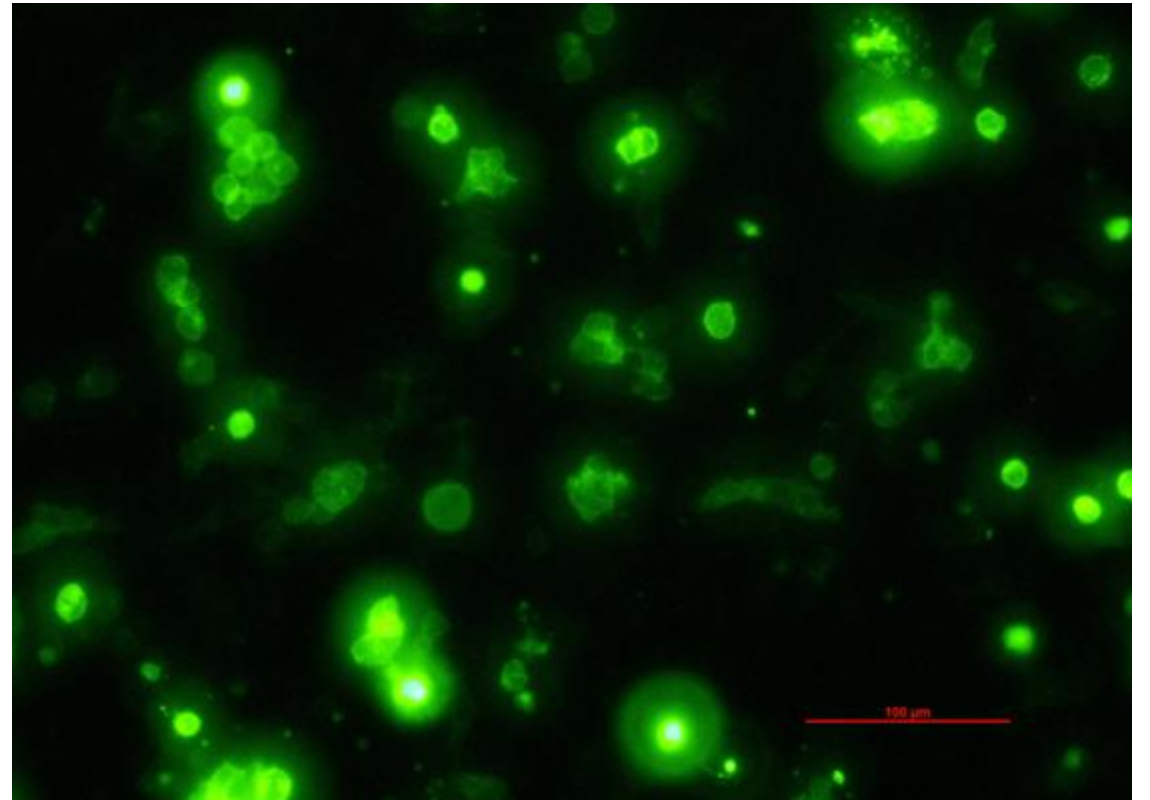
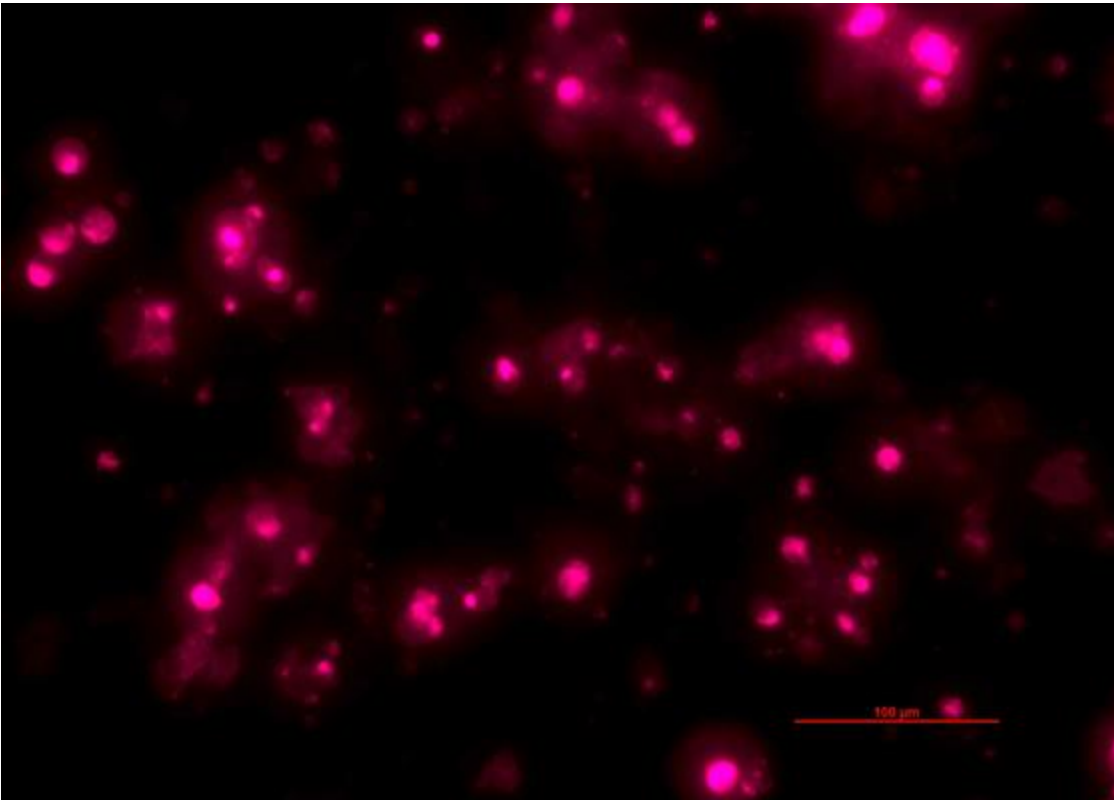
Experiment Timeline (2/3)



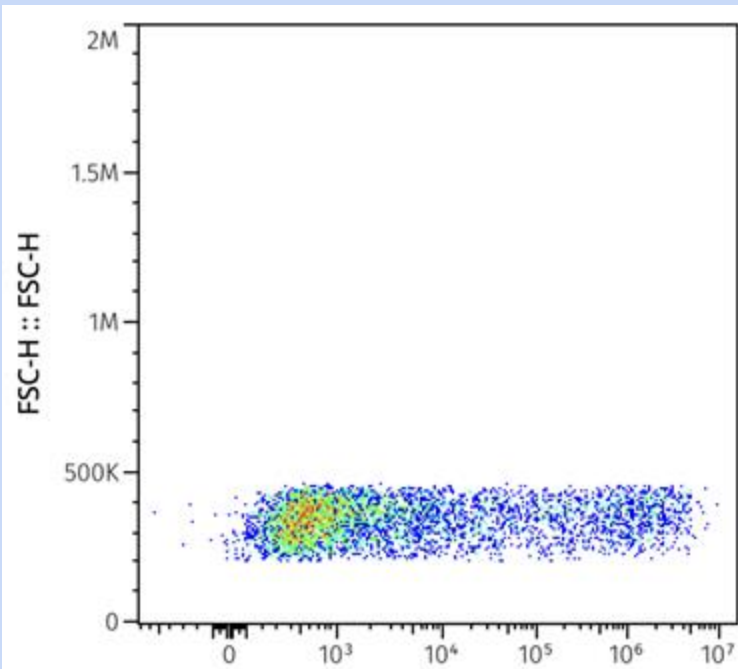
Experiment Timeline (3/3)



Results: Microscope Images

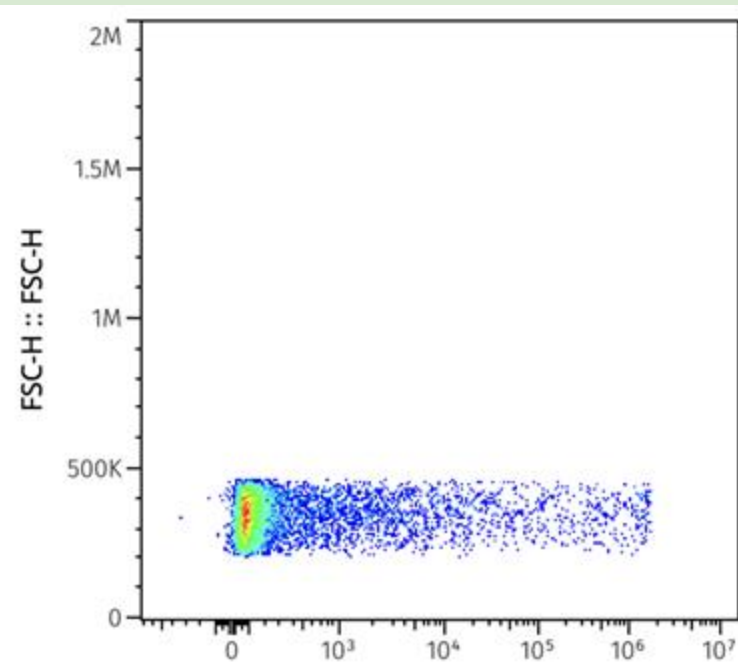


Results: Flow Cytometry Analysis (1/3)



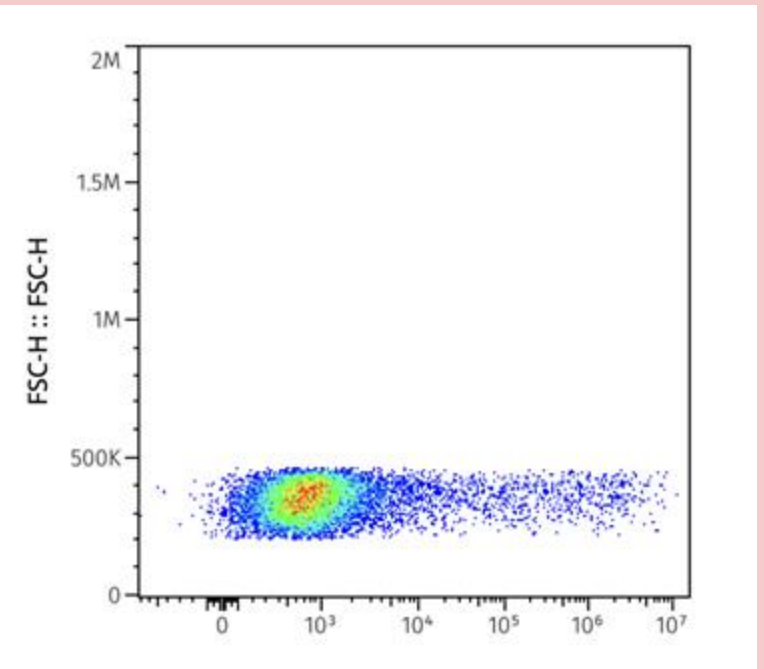
Comp-FL12-A :: TagBFP BV421-A

Sample Name: C14.fcs
Population Name: Singlets
Count: 5722



Comp-FL1-A :: mNeonGreen B525-FITC-A

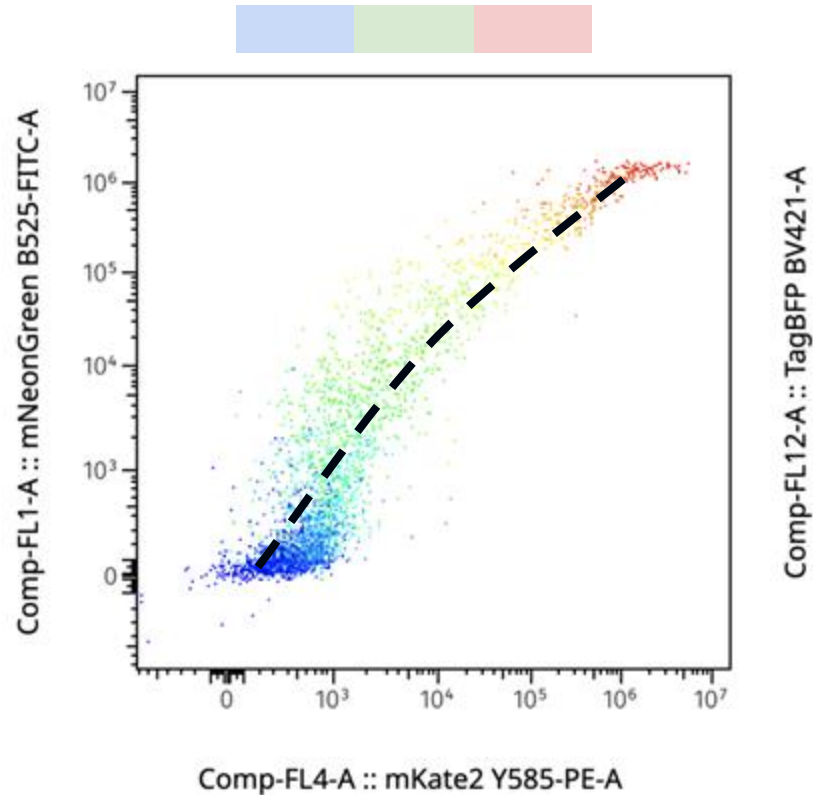
Sample Name: C24 2.fcs
Population Name: Singlets
Count: 10293



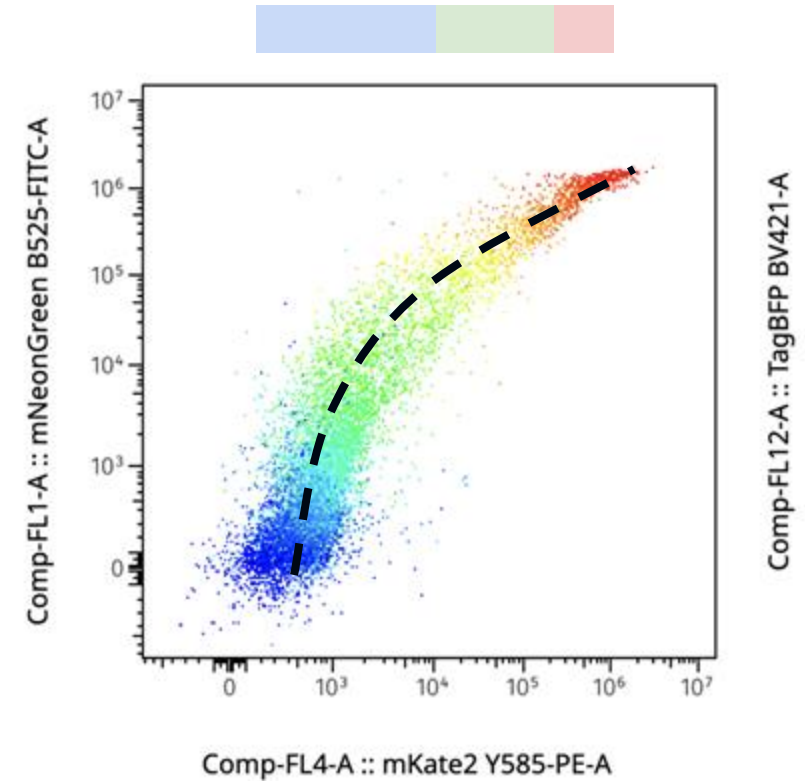
Comp-FL4-A :: mKate2 Y585-PE-A

Sample Name: C34.fcs
Population Name: Singlets
Count: 14461

Results: Flow Cytometry Analysis (2/3)

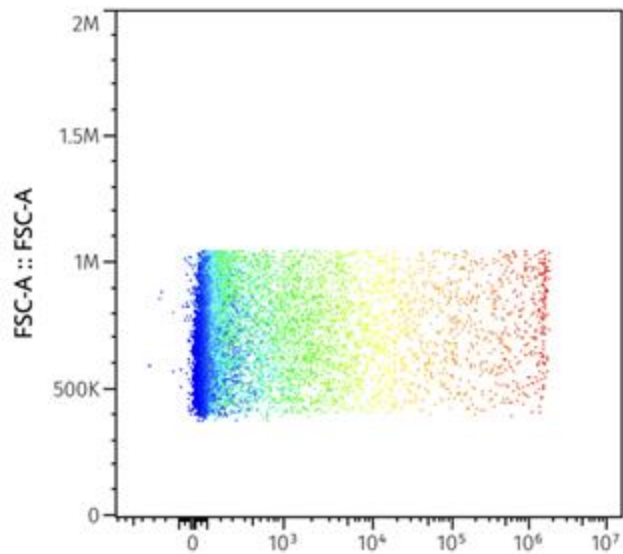


Sample Name: Mix 1.fcs
Population Name: Singlets
Count: 3469



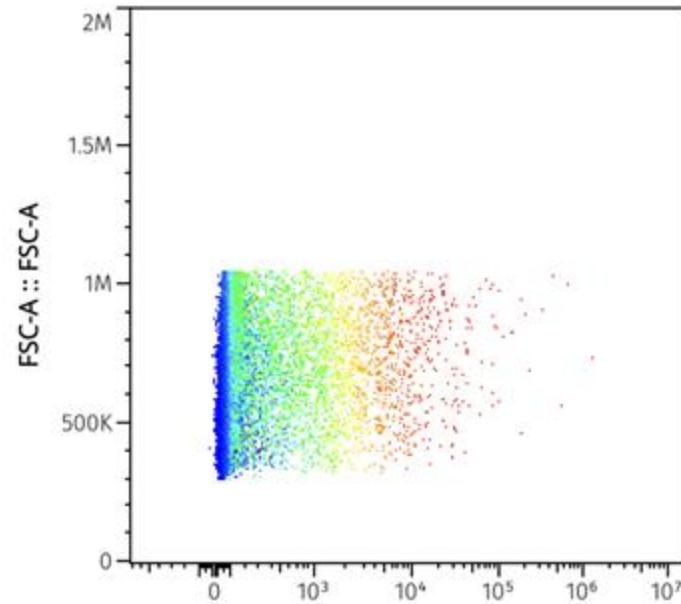
Sample Name: Mix 2.fcs
Population Name: Singlets
Count: 12058

Results: Flow Cytometry Promoter Analysis (3/3)



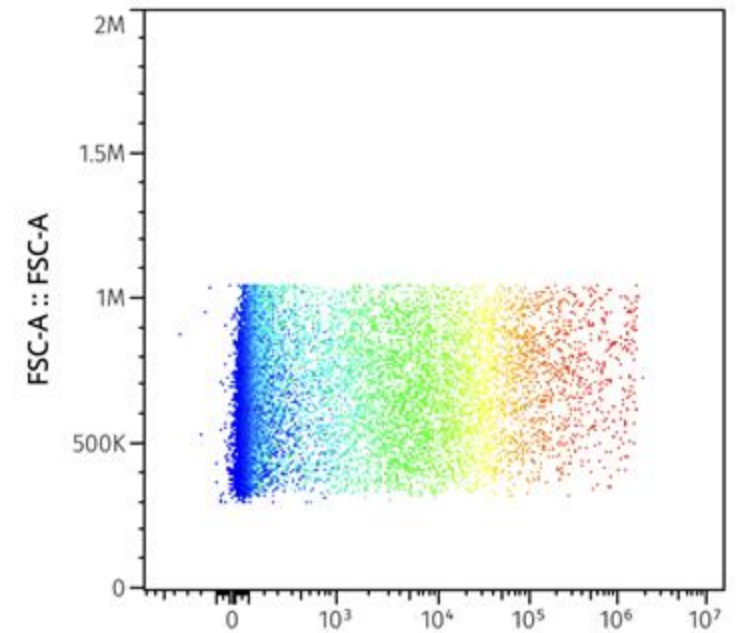
Comp-FL1-A :: mNeonGreen B525-FITC-A

Sample Name: CAG_NG.fcs
Population Name: Singlets
Count: 15317



Comp-FL1-A :: mNeonGreen B525-FITC-A

Sample Name: MNDU3_NG.fcs
Population Name: Singlets
Count: 17486



Comp-FL1-A :: mNeonGreen B525-FITC-A

Sample Name: hEF1a_NG.fcs
Population Name: Singlets
Count: 13885

Conclusion

- 2 of our 5 devices worked successfully; both of which were from our first cycle of production.
- The second wave failed to work at the final stage, however the issue must have been one at a much earlier stage. (Golden Gate)
- It's difficult to make a direct comparison though the promoters were different (4x exposure, double dose, no positive control)

Further Testing

- The device could be tested in other cells to identify their impact
- Proper experimentation with positive & negative controls could provide a better understanding of how different promoters act
- A non-transient transfection method could be used to analyze how the cell takes in the new sequence.

Potential Applications

- Understanding the strength of different promoters can aid in simulating genetic circuits
- A tag that correctly identifies the cell's organelles (eg. mitochondria) could be used to classify cell type (eg. muscle vs fat cell)
- It could be used to understand how organelles respond to environmental stimuli

Thank you

References

1. J. Lin, R. Luo, and L. Pinello, "EPInformer: A Scalable Deep Learning Framework for Gene Expression Prediction by Integrating Promoter-enhancer Sequences with Multimodal Epigenomic Data," bioRxiv, Aug. 2024, doi: <https://doi.org/10.1101/2024.08.01.606099>.
1. S. Seal et al., "Cell Painting: a decade of discovery and innovation in cellular imaging," Nature Methods, vol. 22, pp. 254–268, Dec. 2024, doi: <https://doi.org/10.1038/s41592-024-02528-8>.

